

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 9

The annual rainfall in a remote country varies from 0 to 13 inches and is a random variable with probability density function defined by $f(x) = \frac{13.5 - x}{91}$. Find the following probabilities for the annual rainfall in a randomly selected year.

A. The probability that the annual rainfall is greater than 6 inches.

To find this probability, we will integrate the probability function from $a = 6$ to $b = 13$, since the function is defined on the interval $[0, 13]$.

Separate the integral into two parts and factor the constants from each part. Then find the antiderivatives of 1 and x .

$$\begin{aligned}\int_6^{13} \frac{13.5 - x}{91} dx &= \frac{13.5}{91} \int_6^{13} dx - \frac{1}{91} \int_6^{13} x dx \\ &= \frac{13.5}{91} (x) \Big|_6^{13} - \frac{1}{91} \left(\frac{x^2}{2} \right) \Big|_6^{13}\end{aligned}$$

Now substitute the limits of integration and simplify.

$$\begin{aligned}\frac{13.5}{91} (x) \Big|_6^{13} - \frac{1}{91} \left(\frac{x^2}{2} \right) \Big|_6^{13} &= \frac{13.5}{91} (13 - 6) - \frac{1}{182} (13^2 - 6^2) \\ &\approx 0.3077\end{aligned}$$

B. The probability that the annual rainfall is less than 5 inches.

We set up and solve the integral as in part A. The only difference is the interval over which we integrate. To find this probability, we will integrate the probability function from $a = 0$ to $b = 5$.

Using the new limits of integration, evaluate the probability.

$$\begin{aligned}\int_0^5 \frac{13.5 - x}{91} dx &= \frac{13.5}{91} (x) \Big|_0^5 - \frac{1}{91} \left(\frac{x^2}{2} \right) \Big|_0^5 \\ &= \frac{13.5}{91} (5 - 0) - \frac{1}{182} (5^2 - 0^2) \\ &\approx 0.6044\end{aligned}$$

C. The probability that the annual rainfall is between 5 inches and 10 inches.

Again, set up and solve the integral as in part A, but this time integrate from 5 to 10.

$$\begin{aligned}\int_5^{10} \frac{13.5 - x}{91} dx &= \frac{13.5}{91} (x) \Big|_5^{10} - \frac{1}{91} \left(\frac{x^2}{2} \right) \Big|_5^{10} \\ &= \frac{13.5}{91} (10 - 5) - \frac{1}{182} (10^2 - 5^2) \\ &\approx 0.3297\end{aligned}$$